

# Accuracy Retention Is Not Explanation Retention: Physiological Explanation Drift in Continual EEG Foundation Models

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**Abstract**—Continual adaptation can preserve an EEG decoder’s accuracy while changing the evidence behind its decisions. We study this failure mode in CBraMod across motor-imagery, motor-execution/rest, and sleep-staging tasks. We define channel–frequency reliance by the class-balanced drop in true-vs-best-other classification margin after single-cell spectral occlusion. Before measuring drift, we require subject-level split-half reliability and held-out top-vs-random fidelity. Physiological explanation drift (PED) is then the symmetric cross-checkpoint Jensen–Shannon divergence minus same-checkpoint split noise. Across three task orders, three seeds, and 63 old-task transitions, sequential fine-tuning exhibits substantial PED even when old-task accuracy improves. EWC and DER++ reduce PED relative to matched sequential runs in all 21 comparisons; order–seed-blocked mean reductions are  $-0.122$  and  $-0.080$ , respectively. Yet within-method correlations between relative forgetting and PED remain weak ( $\rho = -0.37, 0.09, -0.21$ ). Thus, performance retention alone does not establish physiological explanation stability in continual EEG foundation models.

**Index Terms**—EEG foundation models, continual learning, explainable AI, attribution stability, catastrophic forgetting

## I. Introduction

EEG foundation models (EEG-FMs) offer a shared representation for heterogeneous clinical and brain–computer-interface tasks [1]. Their downstream lifetime, however, is usually assessed through predictive performance: after learning a new task, does the model still classify an old task correctly? Recent continual EEG work, including Evo-Brain, directly targets this stability–plasticity problem [2]. Accuracy does not reveal whether the model still uses the same physiological evidence.

Explanation drift has been studied in generic continual-learning benchmarks. CLEX showed that similarly accurate models can produce different explanations, and that replay can improve alignment with offline models [3]. SHAP consistency provides a related attribution-stability view on vision benchmarks [4]. Separately, static EEG-FM audits recover task-appropriate spatial or spectral structure [5], [6], while EEG-PRISM maps attribution into physiologically meaningful frequency and source domains [7]. The intersection remains unresolved: does physiologically structured evidence remain stable throughout cross-task continual EEG-FM adaptation?

We ask two questions. RQ1: how much does an old task’s channel–frequency reliance change after a task transition?

RQ2: do methods that reduce performance forgetting also preserve explanations, and is retained accuracy a reliable proxy for explanation stability? Our contributions are:

- a reliability- and fidelity-gated channel–frequency reliance instrument based on held-out spectral occlusion and classification margins;
- a subject-level, noise-corrected PED estimator that separates cross-checkpoint change from same-checkpoint split variability; and
- a matched comparison of sequential fine-tuning, EWC, and DER++ over 27 continual runs, showing that aggregate performance and explanation retention can improve together while transition-level accuracy remains an unreliable proxy for explanation stability.

## II. Reliability-Aware Explanation Drift

### A. Channel–frequency reliance

For task  $i$ , let  $m(x)$  be the true-class logit minus the largest competing logit. Each continuous trial is transformed by rFFT and one channel–band cell is set to zero before inverse rFFT. We use 22 canonical channels and five bands:  $[0.5, 4)$ ,  $[4, 8)$ ,  $[8, 13)$ ,  $[13, 30)$ , and  $[30, 40]$  Hz. Reliance is the decrease in margin caused by the occlusion, averaged first within class, then equally across classes and subjects. Validation rows are deterministically split within each subject/class into attribution-fit and attribution-gate halves. Test rows are never used for attribution.

An attribution instrument must pass two gates before scale. Reliability requires finite split-half cosine similarity, median subject cosine at least 0.70, and at least two thirds of subjects above 0.50. Fidelity selects the top 20% cells on the fit half and requires both margin and subject-balanced-accuracy drops on the gate half to exceed the 95th percentile of 100 equal-cardinality random masks. BCI passes reliability (subject cosines 0.923/0.804) and fidelity. High-Gamma passes with median cosine 0.992 (4/4 subjects above 0.50); its margin fidelity is clear ( $1.801 > 1.594$ ), whereas BA fidelity is narrow ( $0.2563 > 0.2553$ ). Sleep attribution is excluded because it is reliable but fails fidelity.

# Channel-frequency reliance changes despite improved BCI accuracy

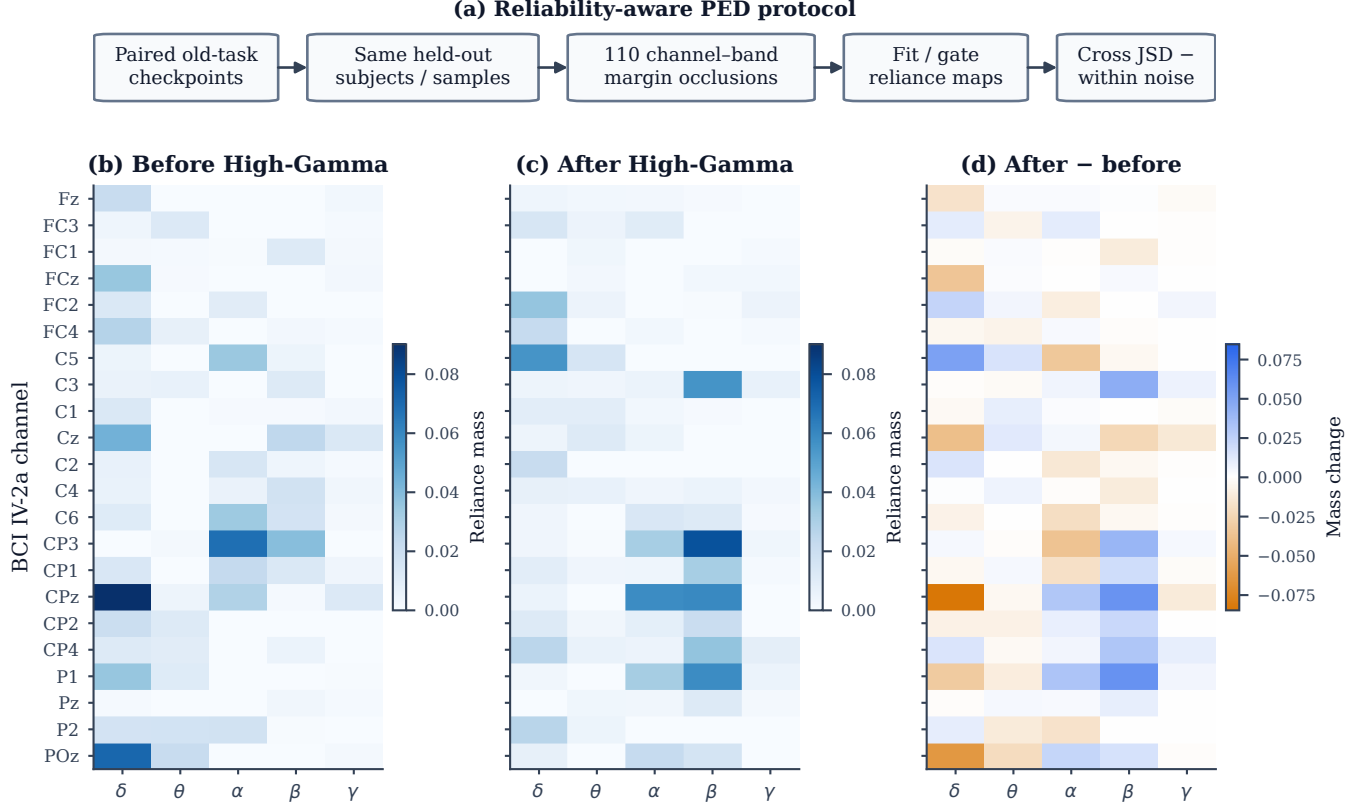


Fig. 1. PED protocol and BCI IV-2a reliance before/after learning High-Gamma. Maps average positive-normalized group reliance over three forward-order seeds; they are not selected from a single favorable run. BCI accuracy improves on average ( $F_{rel} = -0.255$ ), although the channel-frequency distribution changes substantially (PED = 0.243).

## B. Noise-corrected PED

Let  $P_f^b, P_g^b, P_f^a, P_g^a$  be positive,  $\ell_1$ -normalized reliance maps before/after a transition, on the fit/gate halves. For each subject, we define

$$\text{PED} = \frac{1}{2} [\text{JSD}(P_f^b, P_g^a) + \text{JSD}(P_g^b, P_f^a)] - \frac{1}{2} [\text{JSD}(P_f^b, P_g^b) + \text{JSD}(P_f^a, P_g^a)]. \quad (1)$$

Negative estimates are retained. Performance forgetting is  $F_{rel} = (R_{before} - R_{after}) / (R_{before} - R_{chance})$ ; near-chance denominators are reported as invalid rather than clipped.

## III. Experimental Protocol

We fine-tune CBraMod [1] with a shared backbone and a frozen head for each old task. The replacement matrix contains BCI Competition IV-2a (4-class motor imagery, 22 channels, 9 subjects), High-Gamma (4 motor-execution/rest classes, 22 of 128 channels, 14 subjects) [8], and Sleep-EDF Expanded (5-stage sleep classification, two bipolar channels, 78 subjects) [9]. Signals are resampled to 200 Hz, restricted to 0.5–40 Hz, and represented as 1-s patches; High-Gamma uses 4-s trials and Sleep uses 30-s epochs. Subject splits are fixed at 5/2/2, 7/4/3, and 48/15/15 for train/validation/test.

We compare sequential fine-tuning, offline EWC [10] with  $\lambda = 10^5$ , and DER++ [11] with an 8-MiB reservoir (119 items). Hyperparameters are selected once on validation data and reused. Every task receives 2,500 optimizer steps with the final four encoder blocks plastic. Three task orders and seeds {3407, 42, 2026} produce 27 runs. PED is evaluated for old BCI and High-Gamma tasks only, yielding 21 transition cells per method (63 total). Statistical comparisons average transitions within each of nine order-seed blocks, then use a 20,000-replicate block bootstrap and a two-sided sign test. Transition-cell Spearman correlations are descriptive because cells share orders, seeds, and subjects.

## IV. Results

### A. Performance retention does not imply explanation retention

Sequential fine-tuning changes explanations in every direction (Table I). The clearest decoupling is BCI←High-Gamma: BCI performance improves ( $F_{rel} = -0.255 \pm 0.100$ ), while PED remains the largest observed ( $0.243 \pm 0.067$ ; Fig. 1). Therefore, a favorable backward transfer

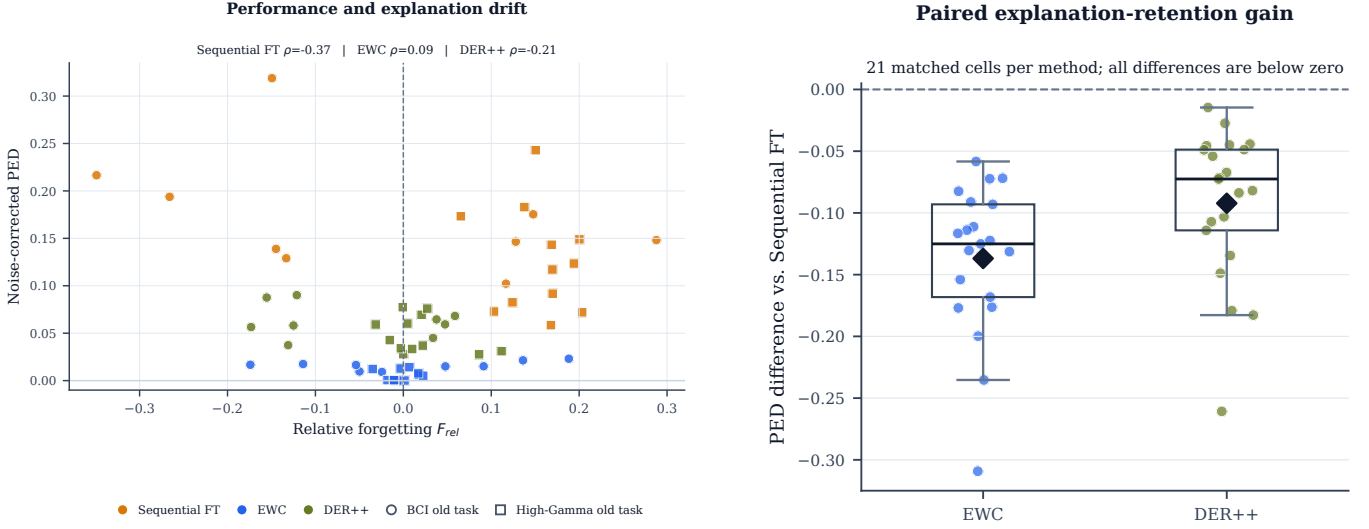


Fig. 2. Performance–explanation alignment. (a) Each point is one matched order–seed–transition cell; color denotes method and marker denotes the old task. (b) Paired PED difference relative to sequential fine-tuning; diamonds show means. Both EWC and DER++ reduce PED in all 21 matched cells.

TABLE I  
Performance forgetting and subject-level noise-corrected PED (mean  $\pm$  SD across matched order–seed transition cells). Lower is better; negative  $F_{rel}$  denotes backward transfer.

| Direction                         | Sequential FT      |                   | EWC                |                     | DER++              |                   |
|-----------------------------------|--------------------|-------------------|--------------------|---------------------|--------------------|-------------------|
|                                   | $F_{rel}$          | PED               | $F_{rel}$          | PED                 | $F_{rel}$          | PED               |
| BCI $\leftarrow$ High- $\gamma$   | $-0.255 \pm 0.100$ | $0.243 \pm 0.067$ | $-0.113 \pm 0.062$ | $0.015 \pm 0.004$   | $-0.070 \pm 0.101$ | $0.052 \pm 0.012$ |
| BCI $\leftarrow$ Sleep            | $0.067 \pm 0.171$  | $0.140 \pm 0.024$ | $0.064 \pm 0.093$  | $0.017 \pm 0.005$   | $-0.053 \pm 0.108$ | $0.069 \pm 0.018$ |
| High- $\gamma$ $\leftarrow$ BCI   | $0.156 \pm 0.036$  | $0.082 \pm 0.020$ | $-0.009 \pm 0.006$ | $0.0003 \pm 0.0002$ | $0.036 \pm 0.051$  | $0.033 \pm 0.006$ |
| High- $\gamma$ $\leftarrow$ Sleep | $0.153 \pm 0.049$  | $0.169 \pm 0.042$ | $0.004 \pm 0.021$  | $0.010 \pm 0.004$   | $0.003 \pm 0.021$  | $0.063 \pm 0.016$ |

score does not establish that the old decision rule was preserved. Across the 21 cells, forgetting and PED are weakly associated within each method: Spearman  $\rho = -0.370$  for sequential fine-tuning, 0.088 for EWC, and  $-0.214$  for DER++. These associations are exploratory and do not support a causal relation.

#### B. EWC and DER++ preserve explanations beyond accuracy

EWC reduces subject-level PED relative to matched sequential runs in 21/21 cells, as does DER++ (Fig. 2). After averaging transitions within each order–seed block, EWC’s mean PED reduction is  $-0.122$  (95% bootstrap CI  $[-0.148, -0.095]$ ); DER++’s is  $-0.080$  ( $[-0.103, -0.058]$ ). Both improve in 9/9 blocks (two-sided sign-test  $p = 0.0039$ ). EWC is strongest: mean PED ranges from 0.0003 to 0.0167 across directions, compared with 0.0333–0.0687 for DER++ and 0.0825–0.2431 for sequential fine-tuning. Thus, the CL methods improve both performance and explanation retention in aggregate, but only direct PED measurement reveals the stability of individual transitions.

#### V. Limitations and Conclusion

The study uses one backbone, three task orders, three seeds, and small BCI and High-Gamma test cohorts. High-Gamma includes overt movement/rest and may contain movement or EMG confounds; it is not treated as pure motor imagery. Its BA fidelity gate passes by only 0.0010. Old-task Sleep explanations are excluded after a predeclared fidelity failure, so physiological claims cover BCI and High-Gamma only. High-Gamma was introduced as a locked replacement after a PhysioNet attribution gate failed, which should be considered when assessing dataset-selection risk. PED measures post-hoc attribution stability, not a biological engram or a causal mechanism of forgetting. We also do not establish alignment with a unique offline explanation.

Within these limits, the evidence is consistent: EWC and DER++ preserve channel–frequency reliance far better than sequential fine-tuning, yet accuracy alone cannot identify whether an individual transition preserves the model’s physiological decision basis. Continual EEG-FM evaluation should therefore report explanation retention alongside performance retention.

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